

Capacity (C) Object Artifact in Dimensional Human Field Theory (DHFT)

DHFT Technical Note

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Abstract

Capacity (C) is a canonical object artifact within Dimensional Human Field Theory (DHFT).

It preserves the invariant identity of Capacity across representations and implementations.

Capacity is fixed as a finite structural threshold and remains non-interpretive and non-substitutable across all admissible representations.

This artifact does not define system-level admissibility or use conditions.

I. Object Designation

Canonical Name: Capacity

Symbol: C

Object Class: Invariant Variable

System Location: Core Mechanical Substrate

Capacity is a primary variable of DHFT and is not derivative or interpretive.

II. Canonical Definition

Capacity (C) is the finite structural threshold of load a system can structurally sustain.

Capacity is structural.

Capacity is amoral.

Capacity is non-interpretive.

Capacity is not defined by effort, intention, or subjective perception.

III. Object Function

The function of Capacity is to define the limit of processable load.

Capacity participates in system determination only in relation to:

- Load (L)
- Boundary (B)

Capacity is not sufficient as an isolated system description.

IV. Invariant Properties

Capacity preserves the following invariant properties:

- Finite: Capacity has a bounded limit
- Structural: Capacity is a system constraint, not a trait
- Non-interpretive: Capacity does not encode effort, intention, or motivation
- Time-dependent: Capacity may vary across time intervals
- Quantifiable: Capacity is representable as a threshold magnitude

These properties are invariant.

V. Prohibited Substitutions

Capacity must not be substituted with:

- resilience
- strength
- grit
- willpower
- endurance
- capability (in narrative sense)

No synonymous, approximate, or domain-specific term may replace Capacity within canonical system use.

VI. Allowed Operations

Admissible operations include:

- comparison with Load (L)
- threshold evaluation (L relative to C)
- temporal variation analysis
- recovery/restoration rate consideration

VII. Non-Admissible Operations

Non-Admissible operations include:

- interpreting Capacity as effort-dependent
- moral or qualitative judgment
- narrative reinterpretation
- treating Capacity as unlimited or expandable without constraint

Such operations constitute non-admissible representation of Capacity and indicate structural drift.

VIII. Relational Constraint

Capacity constrains processable load. Boundary governs load interaction with Capacity.

System state is defined by the relation among L, C, and B.

IX. Representation Constraint

Admissible representation of Capacity must:

- preserve the symbol C
- preserve its finite threshold definition
- preserve non-interpretive use
- preserve relation to L and B

Representations that violate these conditions are non-admissible.

X. Object Admissibility Condition

A representation is admissible as Capacity (C) only if it:

- preserves the canonical symbol (C)
- preserves the canonical definition
- preserves invariant properties
- does not introduce interpretive or domain-specific substitution
- maintains structural relation to L and B

Violation constitutes non-admissible representation and indicates variable-level structural drift.

XI. Completeness Requirement

Capacity must be treated as a complete structural variable.

Partial extraction or isolated use of individual properties does not constitute admissible representation.

XII. Measurement Condition

Capacity is representable as a structural threshold magnitude.

Units may vary but must preserve:

- relative comparison with Load
- threshold conditions
- relation to Boundary

XIII. Object Integrity Condition

Capacity remains the same object across:

- canonical publications
- technical notes
- diagrams
- interfaces
- implementations

Variation in expression does not alter object identity.

Alteration of canonical properties does.

XIV. Verification Reference

Integrity is preserved through:

- canonical DHFT references
- object definitions
- future verification mechanisms

XV. Forward Compatibility

This object definition is substrate-level and remains admissible across future formalizations, including mathematical, computational, and machine-readable implementations.

XVI. Canonical Statement

Capacity (C) is a primary invariant object of DHFT.

It represents the finite structural threshold of load and remains fixed, non-interpretive, and non-substitutable across all admissible representations.

XVII. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All valid system descriptions reduce to Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced at the minimal layer.

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This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

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